Term Project Report and Documentationss

For our term project, we first decided to build a gantry crane because first, it will be easily structured and stabilized based on the materials we have and second, it can support our expected crane size. At early stage of designing, we first used CAD to model out the first version, so that we can start with building up the overall structure and determining the dimensions for other components.

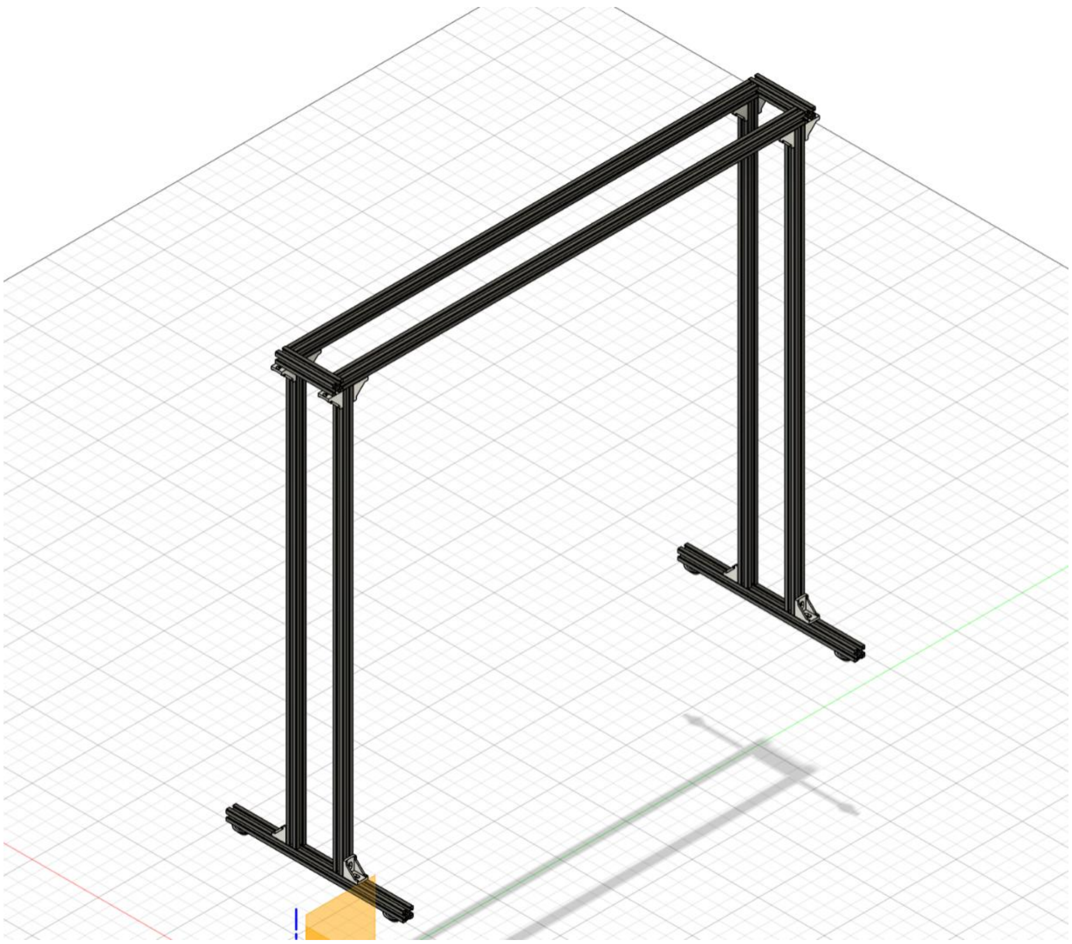

Figure 1: Crane Structure, First Prototype

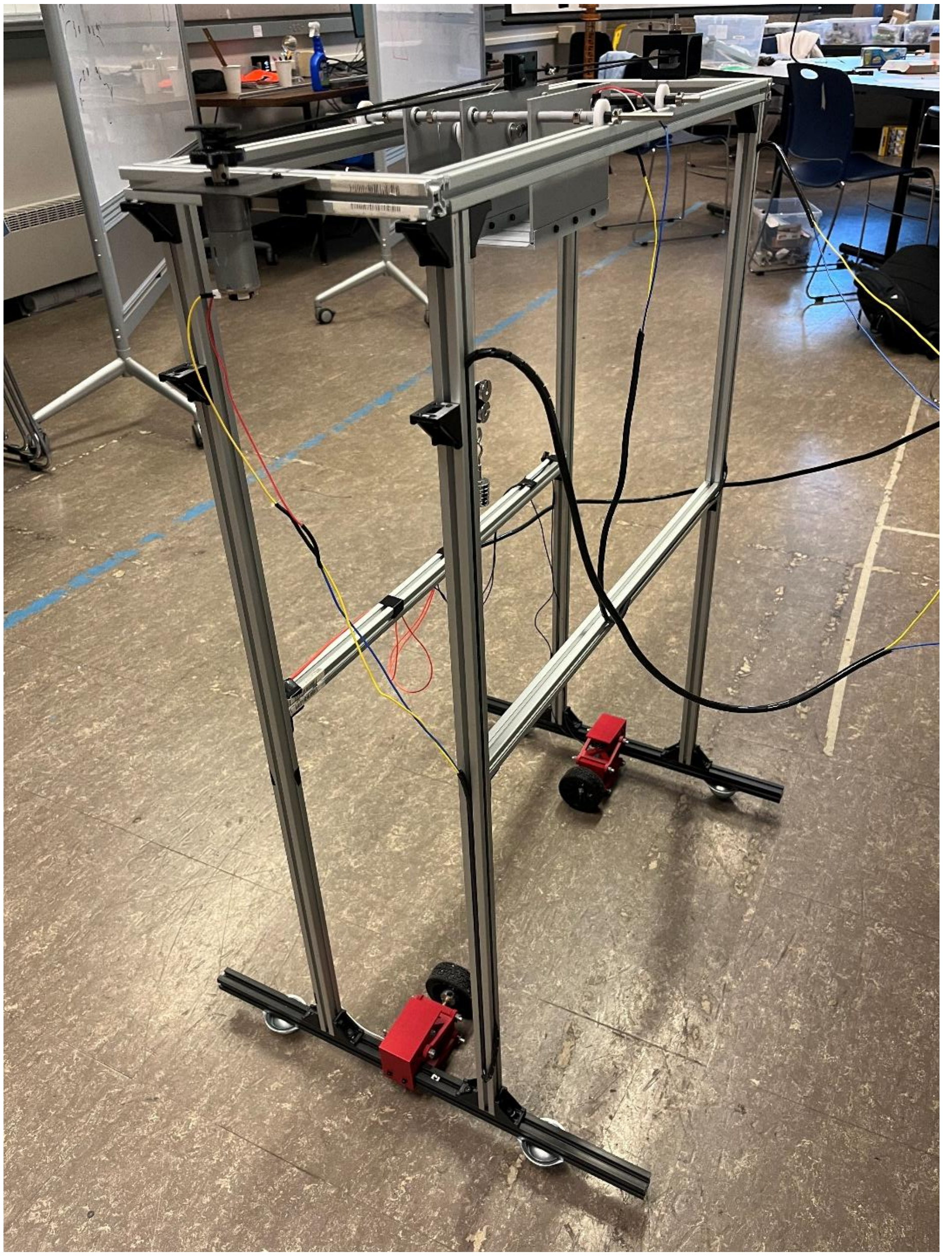

Figure 2: Crane Structure, Final Prototype

We were using the 20mm aluminum extrusions for the structure of our system, so we needed a horizontal movement solution that is compatible with the extrusions and requires minimal additional structural complexity. We observed that the aluminum extrusions had built-in slots on every face; we thought they were the perfect rails for a cart system that could host the vertical-lifting sub-system. The cart mainly consisted of 4 faces, with 3 vertical faces to bear the vertical load from below, and a horizontal plate on the bottom for support. Small holes were made at the intersection point to ensure a strong connection.

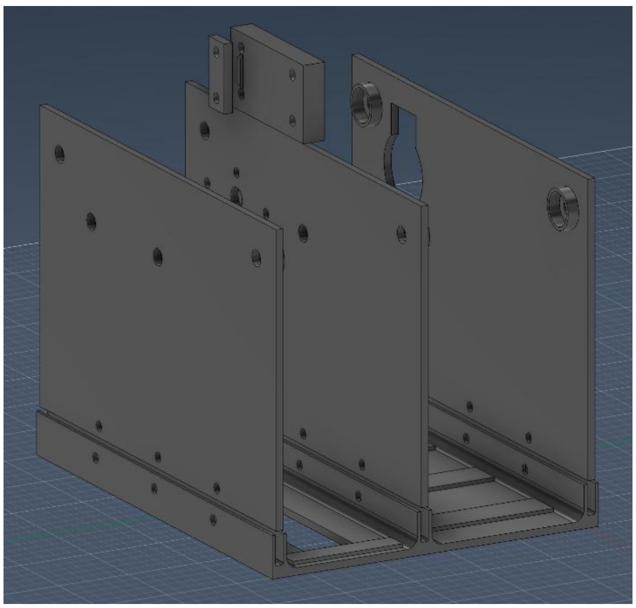

Figure 3: Motor-Pulley Housing, View 1

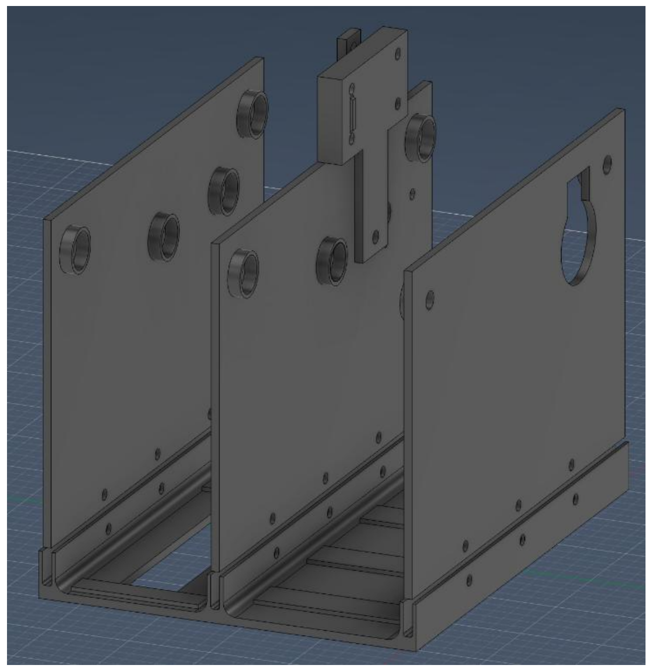

Figure 4: Motor-Pulley Housing, View 2

Holes were made where the axials go through, which may include motor, winches, and 5mm shafts. Shaft collars were used with the 5mm shafts, which, when combined with custom-made 5.3mm 3D printed spacers, provide axial support to the 3 vertical faces. Various sizes of spacers were made to best fit the axials.

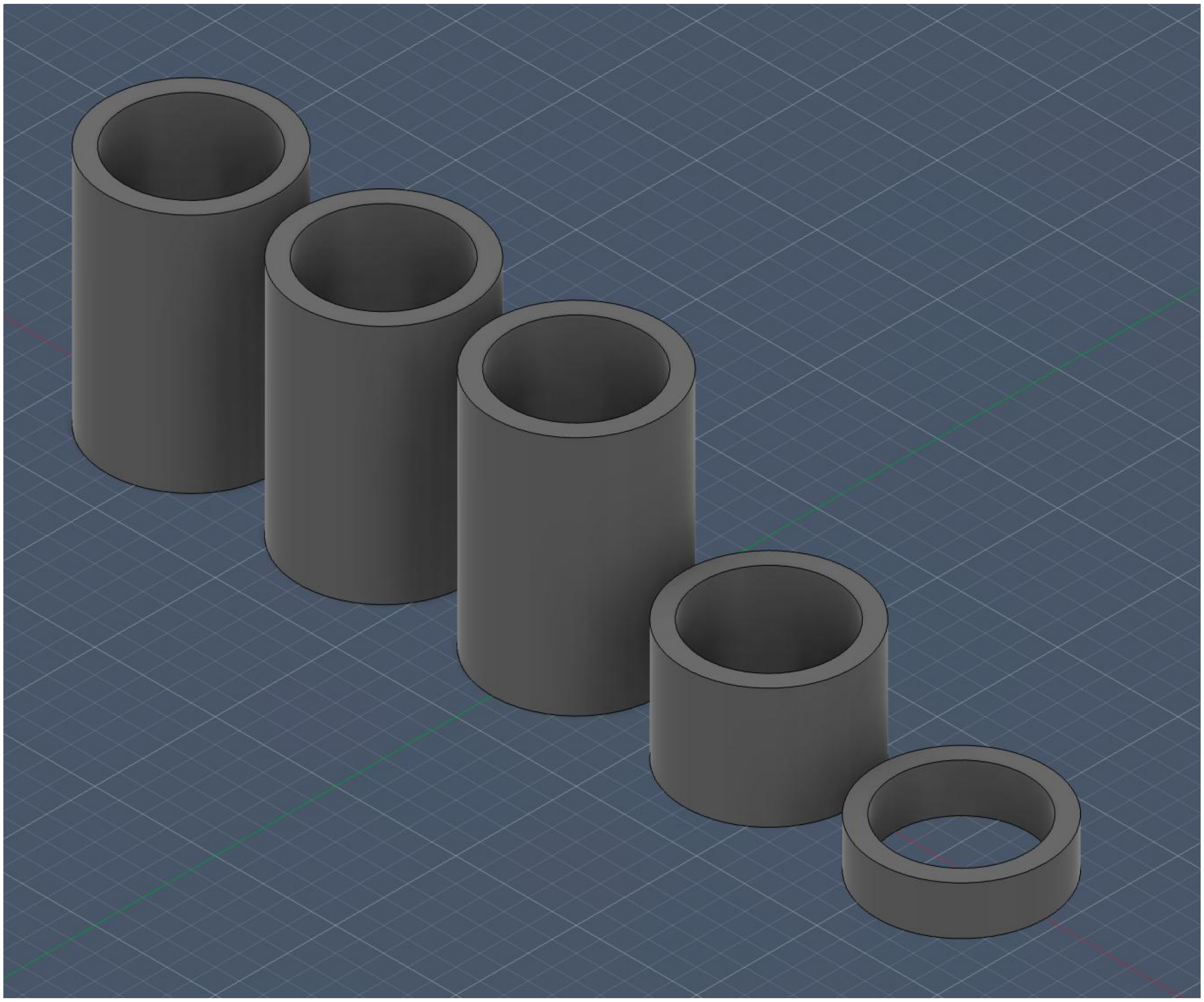

Figure 5: Spacers

A 12V motor with planetary speed reduction was used to power the winch. A 30lb fishing line was used to lift the load.

The fully assembled cart system is as follows:

Figure 6, 7, and 8: Cart System

Custom winches were printed to work with a 5mm shaft, a cone-shaped design was used to guide the fishing line. Nylon wire was used to pull the cart along the slots on the aluminum extrusion. One side of the horizontal pulley used custom-designed housing, which deflects under tension to allow the string to be properly strung.

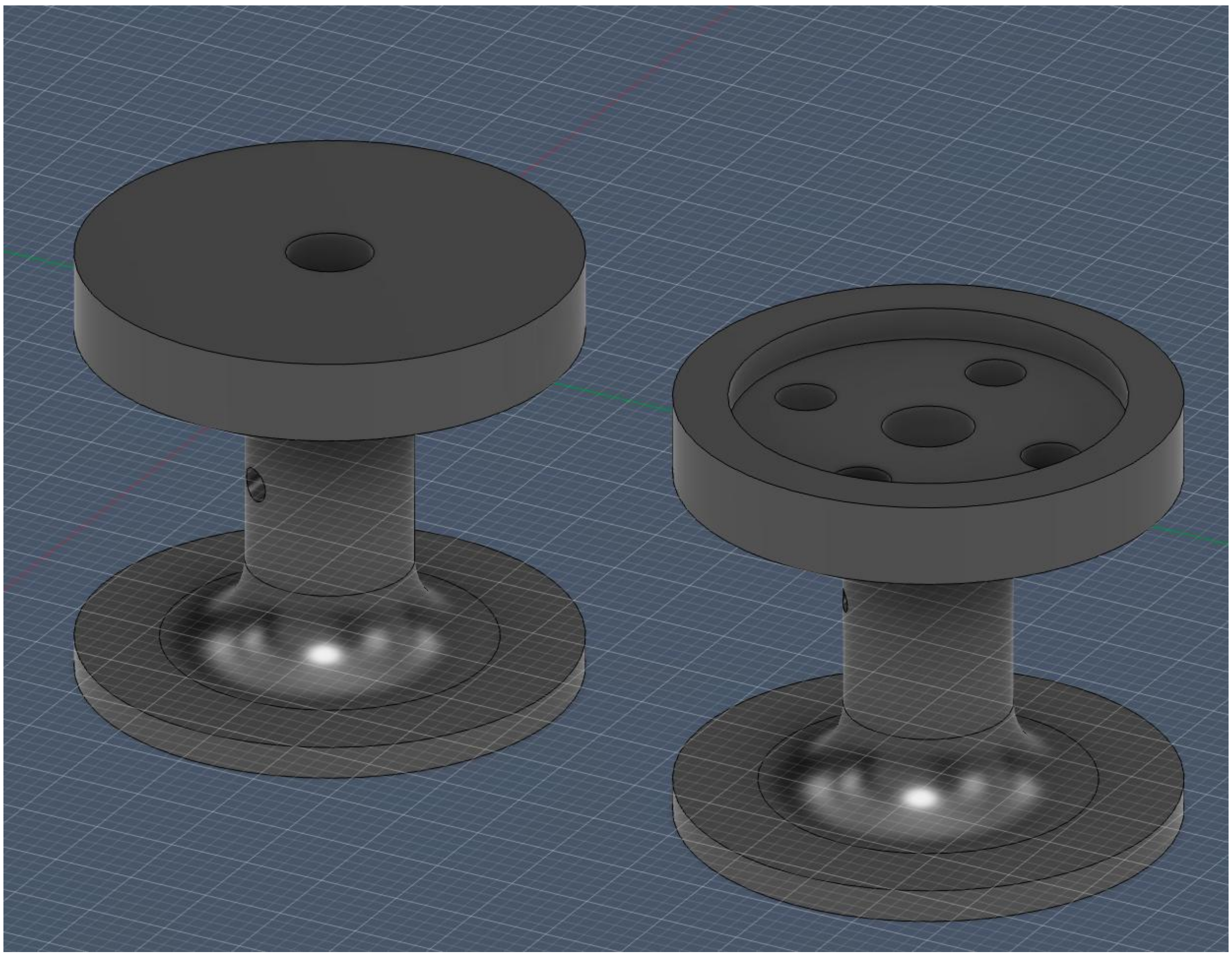

Figure 9: Pulleys

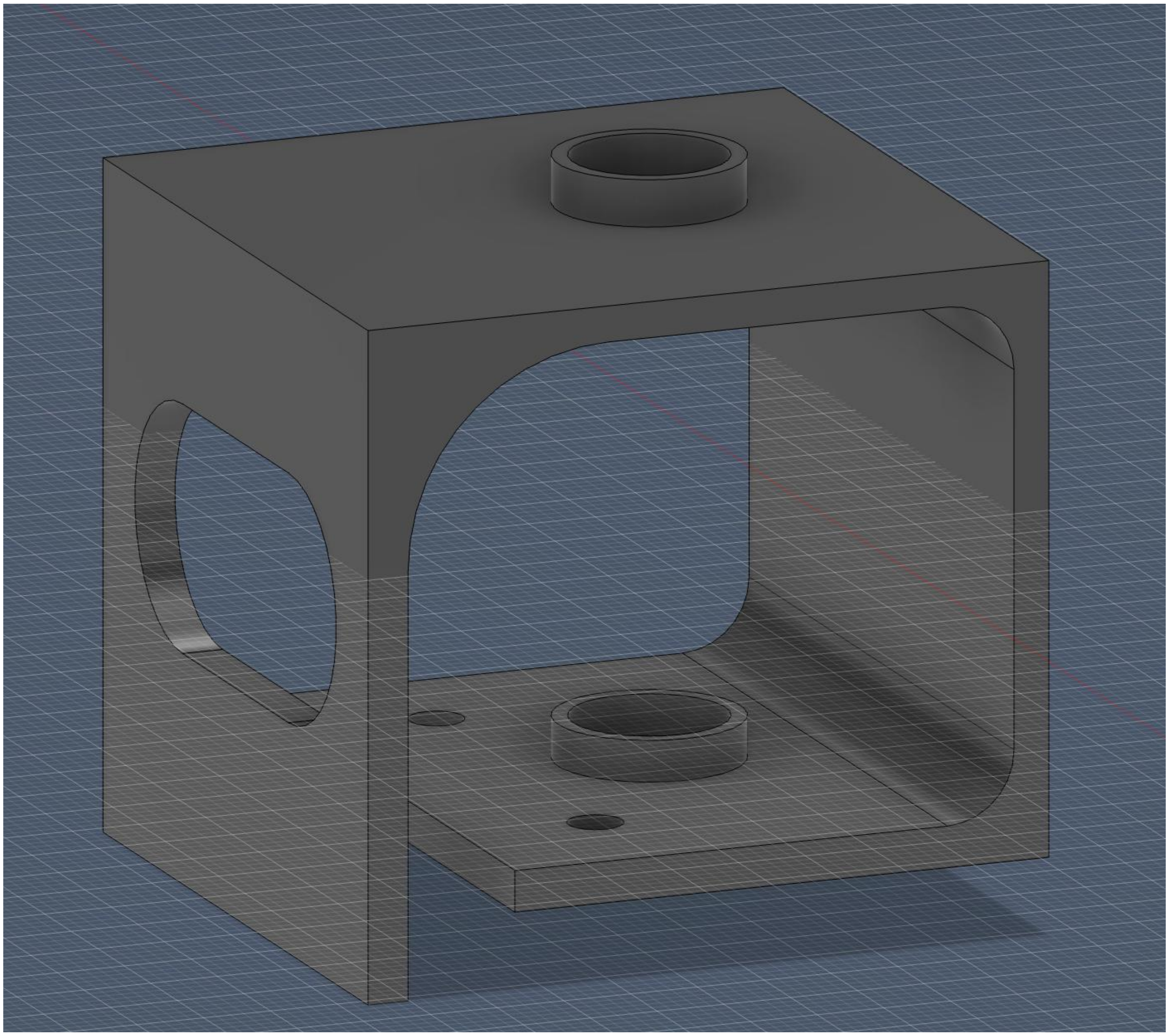

Figure 10: Pulley Housing

A special design was used on the horizontal pulley set, as one side was a regular, smooth shaft pulley, while the pulley mounted to the actuator used a different design, where the shaft had spikes on the surface for a better grip on the string.

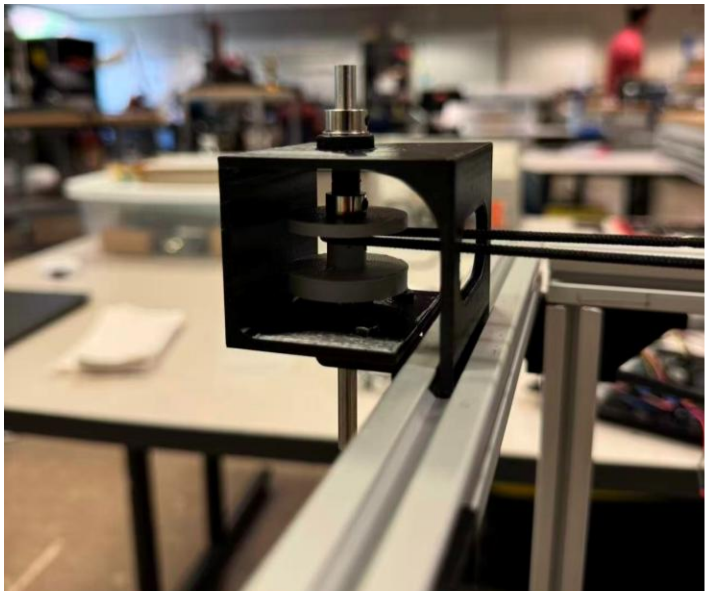

Figure 11: Actual Pulley Housing

A special design was used on the horizontal pulley set, as one side was a regular, smooth shaft pulley, while the pulley mounted to the actuator used a different design, where the shaft had spikes on the surface for a better grip on the string.

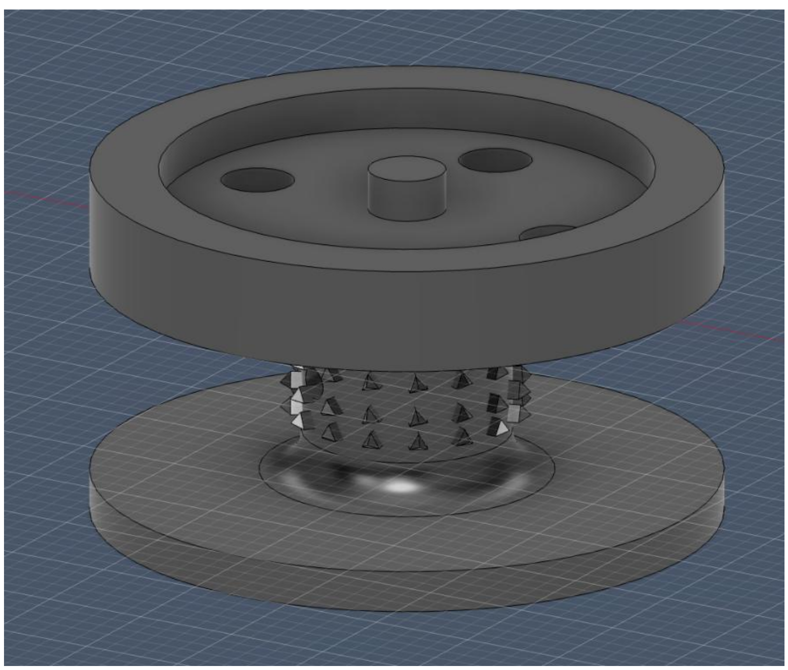

Figure 12: Special Pulley Design

On a side note, all the pulleys are connected using a 5mm flange, with 4 screws directly installed through the disk on the pulley. Since we had 4 carrier bearings on each of the corners of the base, which makes it an undetermined system, it was challenging to design a drive system that put as much weight from the structure as possible onto the actuating wheels while keeping all the rest of the bearings in contact with the ground. A parallel 4-bar suspension system was then developed and implemented. Off the shelf 65mm diameter rubber wheels were used in the final assembly, and a 12V speed-reduced motor was installed on each side of the suspension cage. Springs were installed between the tower and the upper control arm to press the wheel to the ground with consistent force.

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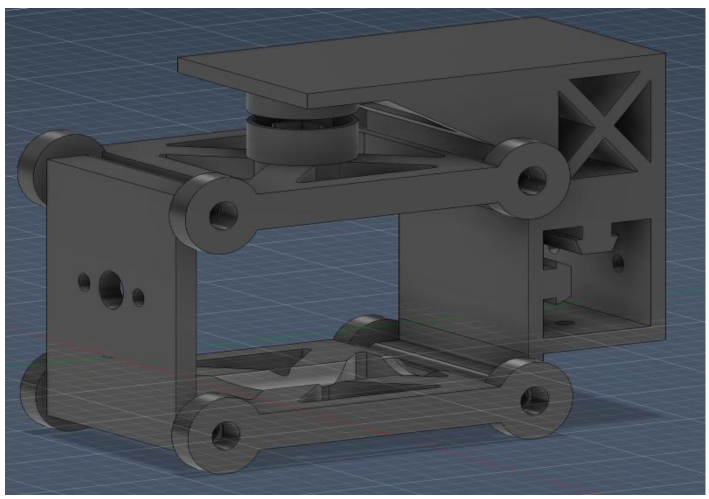

Figure 13: Wheel Suspension System in Fusion

Figure 14: Actual Wheel Suspension System

As for the drive base controls, a differential controller was used to control the turning and translational motion of the whole system. An Arduino controller was used to perform the switching, and a L298N motor driver was used to perform the motor actuation, as the motor and Arduino run on different voltages, two battery packs were used. The detailed circuit is shown below:

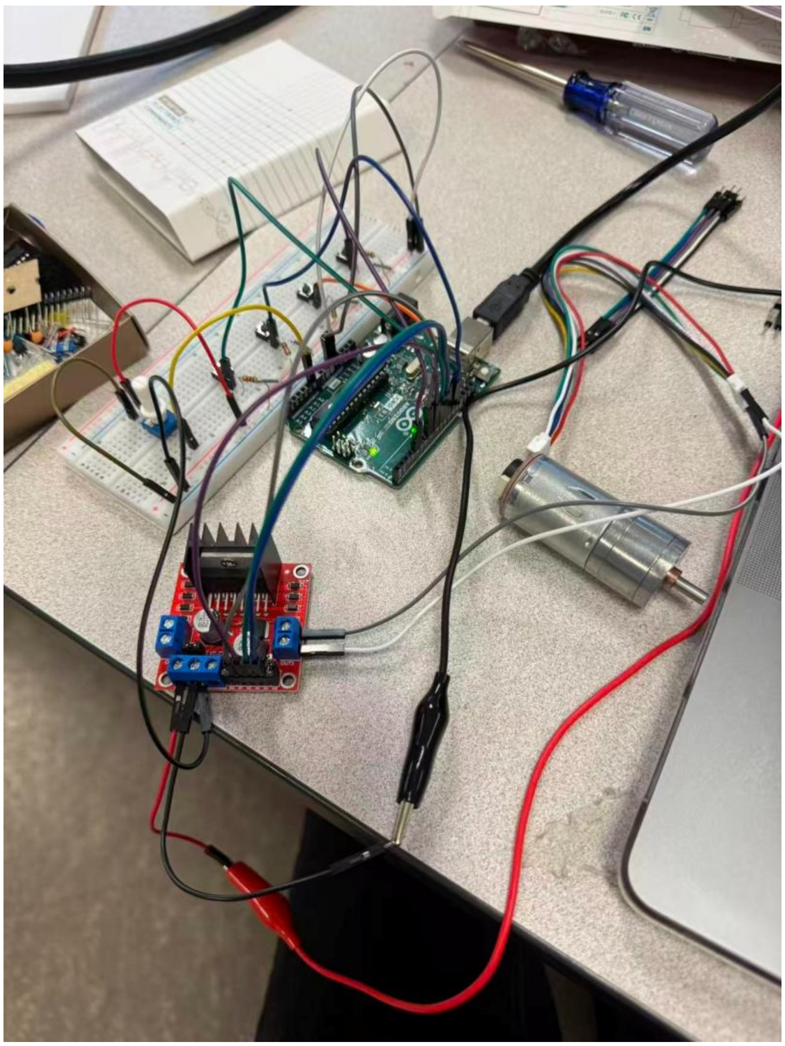

Figure 15: Circuit for Wheel Control

For the controls of the motors for vertical and horizontal movement, two DPDT switches and two motor controllers were used to switch polarities, thus, to change the movement direction, and control the motor speed. A total voltage of 12V was applied to run this controller with two AA batteries and one 9V battery in series. The detailed circuit is shown below:

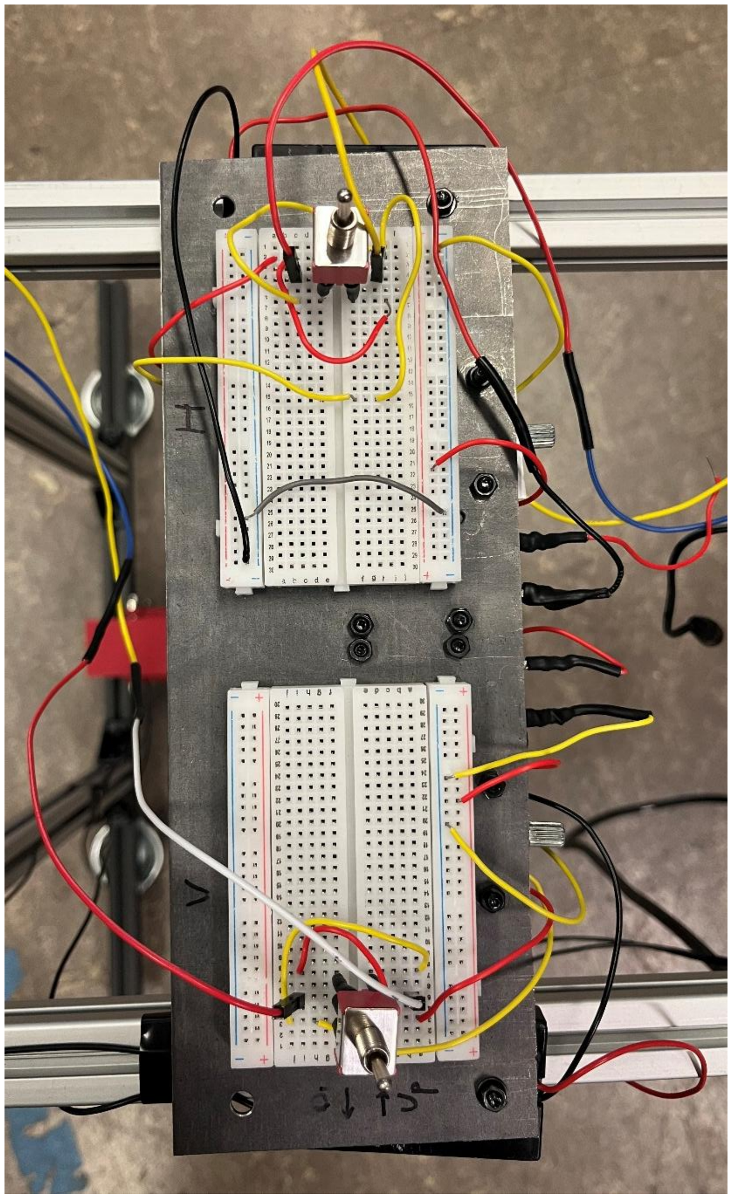

Figure 16: Circuit for Vertical and Horizontal Movement Control

The final configuration of our crane system looks as follows:

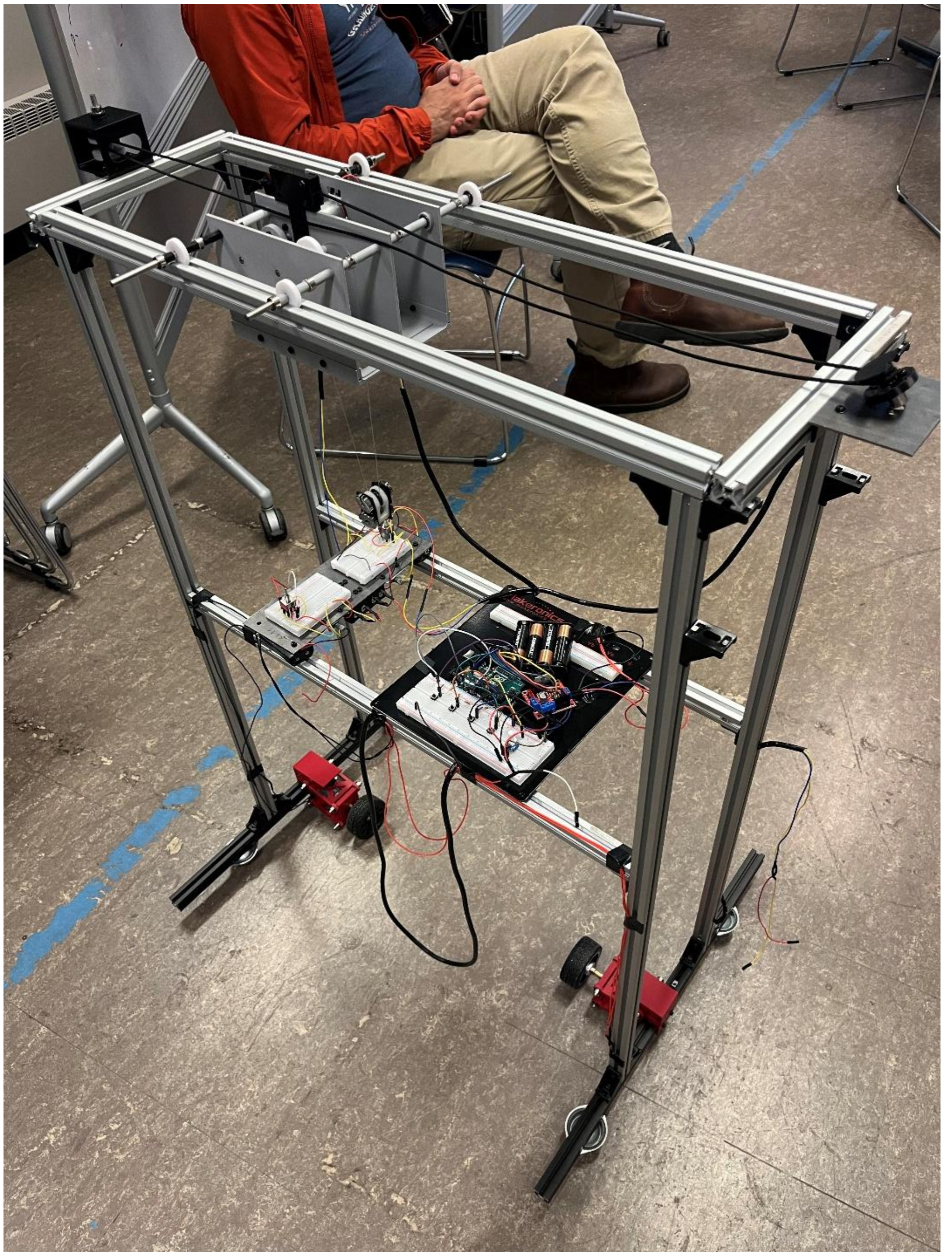

Figure 17: Crane Final Configuration

Appendix

Arduino codes for wheel control:

sketch_may21b.ino

```
// -----------------------------------------------
// Two DC motors with L298N
// 4 buttons: direction control
// 10k potentiometer: shared PWM speed control
// External pull-down resistors on buttons
// -----------------------------------------------


const int ENA = 5;   // PWM for Motor 1
const int ENB = 6;   // PWM for Motor 2

// L298N direction pins
const int IN1 = 7;
const int IN2 = 8;

const int IN3 = 9;
const int IN4 = 10;

// Buttons with external pull-down resistors
const int M1_FWD_BUTTON = A1;
const int M1_REV_BUTTON = A2;

const int M2_FWD_BUTTON = 4;
const int M2_REV_BUTTON = 11;

// Potentiometer center/wiper pin
const int POT_PIN = A0;


// power ranges from -255 to +255.
// Positive = forward, negative = reverse, zero = stop.
void setMotor(int enablePin, int dirPin1, int dirPin2, int power) {
  power = constrain(power, -255, 255);

```

sketch_may21b.ino

```
  if (power > 0) {
    digitalWrite(dirPin1, HIGH);
    digitalWrite(dirPin2, LOW);
    analogWrite(enablePin, power);
  }
  else if (power < 0) {
    digitalWrite(dirPin1, LOW);
    digitalWrite(dirPin2, HIGH);
    analogWrite(enablePin, -power);
  }
  else {
    digitalWrite(dirPin1, LOW);
    digitalWrite(dirPin2, LOW);
    analogWrite(enablePin, 0);
  }
}


void setup() {
  pinMode(ENA, OUTPUT);
  pinMode(ENB, OUTPUT);

  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);

  // External pull-down resistors:
  // not pressed = LOW, pressed = HIGH
  pinMode(M1_FWD_BUTTON, INPUT);
  pinMode(M1_REV_BUTTON, INPUT);
  pinMode(M2_FWD_BUTTON, INPUT);
  pinMode(M2_REV_BUTTON, INPUT);

  setMotor(ENA, IN1, IN2, 0);
  setMotor(ENB, IN3, IN4, 0);
```

sketch_may21b.ino

```
}


void loop() {
  int potValue = analogRead(POT_PIN);        // 0-1023
  int pwm = map(potValue, 0, 1023, 0, 255);

  if (pwm > 0 && pwm < 70) {
    pwm = 70;
  }

  bool m1Forward = digitalRead(M1_FWD_BUTTON) == HIGH;
  bool m1Reverse = digitalRead(M1_REV_BUTTON) == HIGH;

  bool m2Forward = digitalRead(M2_FWD_BUTTON) == HIGH;
  bool m2Reverse = digitalRead(M2_REV_BUTTON) == HIGH;

  int motor1Power = 0;
  int motor2Power = 0;

  // Both pressed or neither pressed = stop
  if (m1Forward && !m1Reverse) {
    motor1Power = pwm;
  }
  else if (m1Reverse && !m1Forward) {
    motor1Power = -pwm;
  }

  if (m2Forward && !m2Reverse) {
    motor2Power = pwm;
  }
  else if (m2Reverse && !m2Forward) {
    motor2Power = -pwm;
  }

  setMotor(ENA, IN1, IN2, motor1Power);
  setMotor(ENB, IN3, IN4, motor2Power);
}
```